Environmental Data Science: Homework #8
Random Forests
Due: Before Class on 10/21/26

Tree-based model regression in python – NOAA Buoys

We'll be using data from a NOAA buoy off the coast of California. A link to the data descriptions can be found here: www.ndbc.noaa.gov/station_page.php?station=46222

Data are posted on blackboard. [The file is whitespace delimited, so pd.read.csv will not work by default.]

Using these data, train models to predict wave height as a function of dominant wave period, average wave period, the direction from which the waves at the dominant period are coming, and sea surface temperature

1. Fit a decision tree regressor and random forest regressor with 25 trees. Compare with a baseline linear model of your choice. Which model performs best? Report performance on the training, validation, and test dataset.
2. What cost function are you using for each model?
3. What (if any) hyperparameters did you optimize?
4. If you redo your train/validation splits, do you get the same answers?
5. Train a random forest model on this prediction task with a 2, 10, 25, 50, 100, 250, 500, & 1,000 trees. Plot the training and validation error as a function of number of trees.
6. Do you have evidence of overfitting to the training dataset with 1,000 trees?